

SYEED ARSHAD

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Summary

B.Tech ECE student (CGPA 8.52/10, Class of 2027) actively preparing for software engineering roles in the tech industry, with hands-on experience building AI-powered, full-stack software: RAG (Retrieval-Augmented Generation) pipelines, multi-agent systems, REST APIs, and production-style backends in Python/FastAPI and React/TypeScript. Google AI Professional Certificate holder with additional hands-on experience in signaling systems and hardware diagnostics from two engineering internships.

Experience

Signals and Telecommunication Intern — South Central Railways

Jul 2025 – Sep 2025

Andhra Pradesh, India

- Analyzed fault-detection workflows and interlocking logic for a Network Failure Detection System across 4 operational departments, improving alert-generation reliability for safety-critical infrastructure.
- Audited real-time signaling and monitoring pipelines, identifying systemic gaps in live railway control architecture.

Electronics Intern — Frontline Electronics Pvt. Ltd.

Jun 2025

Telangana, India

- Diagnosed hardware faults in aviation warning lighting systems via PCB-level root-cause analysis; performed PCB fabrication, SMT inspection, and quality-control workflows.

Projects

StudyFlow AI — AI-Powered Student Productivity Platform

Jun 2026 – Aug 2026

Electron, JavaScript, React, TypeScript, Vite, Python, FastAPI, SQLite, PostgreSQL, GitHub Actions

- Built and shipped a full-stack, cross-platform desktop productivity app (Electron + JavaScript, FastAPI backend, SQLite/PostgreSQL) covering task management, goal roadmaps, study planning, focus mode, exam prep, notes, analytics, wellness tracking, and an AI study coach.
- Implemented a multi-provider AI management system for personalized study assistance, and a responsive marketing site (React, TypeScript, Vite).
- Added CI workflows and automated testing across backend and desktop components, reaching 240+ passing tests; open-sourced the project with full documentation, contribution guidelines, and a security policy.

JASS AI — AI-Powered Personal Assistant Platform

Mar 2026 – Present

Python, FastAPI, React, FAISS, RAG, WebSockets, JWT, SQLAlchemy, Groq, Gemini, Ollama, Whisper

- Engineering a hybrid Retrieval-Augmented Generation pipeline combining FAISS dense vector search with BM25 sparse retrieval, query expansion, and context compression for long-term memory.
- Building a JWT-secured FastAPI backend with SQLAlchemy and WebSockets, plus a multi-provider LLM routing layer across Groq, Gemini, and Ollama with real-time voice input via Whisper.

AI Email Assistant

May 2026 – Jun 2026

Python, Gmail API, OpenRouter, SQLite, Streamlit, Telegram Bot API, OAuth 2.0

- Automated email classification into 5 categories, summarization, priority scoring, and reply drafting using Gmail OAuth 2.0 and OpenRouter LLMs.
- Engineered a human-in-the-loop decision layer routing personal/commitment emails through a Telegram approval workflow (Send, Edit, or Ignore), with a Streamlit analytics dashboard and SQLite audit logging.

Line Detection & Obstacle Avoidance Robot Car

Feb 2026 – Mar 2026

- Built an autonomous ESP32 robot (IR + ultrasonic sensors, C++) with motor control and sensor-fusion logic for dynamic path correction.

Technical Skills

- Languages:** Python, JavaScript, TypeScript, C++
- AI/ML:** RAG (Retrieval-Augmented Generation), Vector Databases, FAISS, BM25, Generative AI, LLMs, Prompt Engineering, Groq API, Gemini, Ollama, OpenRouter
- Web & Backend:** FastAPI, REST APIs, API Development, WebSockets, React, Node.js, HTML, CSS
- Databases:** MySQL, SQLite, PostgreSQL, SQLAlchemy | **Auth & Security:** JWT, OAuth 2.0
- Tools & Platforms:** Git, GitHub, GitHub Actions, CI/CD, VS Code, Arduino IDE, LTspice, Streamlit, Electron
- Domains:** AI Systems, Full-Stack Development, Software Engineering, Embedded Systems, Signal & Telecommunication

Education

B.Tech, Electronics and Communication Engineering

2023 – 2027

Vel Tech Rangarajan Dr. Sagunthala R&D Institute — Chennai, India | CGPA: 8.52/10 | Class XII: 82.6% | Class X: 99.7%

Certifications & Achievements

- Google AI Professional Certificate — Google via Coursera, May 2026
- JavaScript Certification — Udemy, Oct 2025 | MySQL Certification — Udemy, Dec 2025
- HackVega 2.0 — Final Round Qualifier, nationwide hackathon (47,000+ registrations), by MyCareernet & HirePro
- IEEEExtreme 18.0 — Global Rank 1788, India Rank 556, University Rank 15 (IEEE's 24-hour international programming contest), Oct 2024